

Second-Order Multidimensional Guiding Vector Field for Parametric Path-Following Under Holonomic Constraints

Havish Sripada and Baron Henderson
email: info@pedropathing.com

Abstract

Autonomous path-following in higher-dimensional workspaces is a crucial yet challenging task for modern industrial applications. However, traditional geometric path-followers fail to accommodate robots requiring actuator control via acceleration-level commands. This paper proposes a lightweight guiding vector field (GVF) framework that commands second-order dynamics to support holonomic path-following over the Special Euclidean group $SE(n)$. Defined as a map from the first to the second tangent bundle of this configuration space, the GVF balances four decoupled components: an attractive acceleration field, a skew-projected heading field, a feedforward-based centripetal acceleration field, and a kinematically-modeled tangential drive field. Using techniques such as smooth mollifiers to extend the closest point projection, the paper also tackles the GVF-based path-following of parametrically-defined curves, another problem that has traditionally proved difficult to solve. Through rigorous mathematical analysis, this paper proves that the algorithm retains the geometric guarantees of traditional GVF algorithms, such as smoothness, path invariance, and stable asymptotic convergence to the path. This architecture serves as the basis of the Pedro Pathing motion-planning and path-following library.

1 Introduction

Autonomous robotics path-following has become increasingly relevant in modern industry. With the complexity of robotic tasks, unmanned vehicles must rapidly follow and converge to reference paths. Higher-dimensional workspaces intensify the difficulty in developing a strong path-following algorithm, as developing a unified theory supporting the geometries of manifold embeddings in ambient Euclidean spaces of arbitrary dimension proves to be a challenging task [Yao+23]. The necessity to support acceleration-based commands instead of velocity-based commands further amplifies the challenges of constructing such path-followers, necessitating the analysis of second-order dynamics.

Guiding vector fields (GVFs) have long been recognized as a solid approach to robotic path-following [Nel+07]. As geometric path-followers, they boast robustness, optimality or near-optimality, and time-independence [Sam92]. However, unlike most other geometric path-followers, such as Pure Pursuit or Stanley controllers, GVFs can easily be generalized to higher-dimensional ambient spaces [Gon+10]. On the other hand, guiding vector fields often struggle to follow curves embedded in Euclidean space through parameterizations, instead preferring implicit representations of such 1-manifolds. More critically, geometric path-followers are typically limited in their inability to support robots with acceleration-based commands, an often detrimental obstacle for robots with certain actuator models [Sni09]. For this reason, dynamic path-followers are often preferred under such conditions [Bor+05]. Still, they typically require an explicit robot kinematic model along with significantly more system identification, and demonstrate increased computational cost and sensitivity to noise [Pad+16]. Computational complexity also scales exponentially in higher-dimensional spaces.

This paper proposes a simple and dynamic GVF framework that supports path-following along the Special Euclidean group $SE(n)$. Extending geometric path-following to the commanding of second-order dynamics, the paper leverages techniques of model-based dynamic path-followers to develop a GVF capable of producing acceleration-based commands. The proposed second-order GVF retains the robustness and geometric guarantees of standard GVFs, such as smoothness, along with global asymptotic convergence to and the invariance of the reference path. The remainder of the paper is as follows: Section 2 describes the problem and assumptions in detail, Section 3 develops the GVF algorithm, and Section 4 proves various important properties critical to the algorithm.

1.1 Pedro Pathing

This paper details the underlying algorithm behind the open-source motion-planning and path-following library Pedro Pathing [Ped24b], developed by an Oregon-based nonprofit under the same name. The algorithm was originally designed by developers Anyi Lin and Harrison Womack, along with the rest of FIRST Tech Challenge Team 10158 Scott’s Bots, as a motion planning library for the FIRST Tech Challenge competition, named Pedro Pathing, after their 2023-24 season’s robot. The library was later developed by Logan Nash and Aarsh Mittal, while the authors of this paper are the current developers of the library. At the time of writing this paper, the library has amassed thousands of users on a global scale. Pedro Pathing also develops software for trajectory visualization, command-based logic frameworks, and control for unmanned aerial vehicles [Ped24a].

2 Problem Formulation

2.1 Notation

Given a tangent vector v on a normed vector space, we use the shorthand $\hat{v} := \frac{v}{\|v\|}$. Unless otherwise specified, vectors are denoted by bolded lowercase letters (e.g., $\mathbf{x}$), matrices are denoted by italicized uppercase letters (e.g., R), and scalars are generally denoted by italicized lowercase letters (e.g., v). For a topological set A , the symbol ∂A denotes the boundary of A whereas the symbol A° denotes the interior of A . The norm $\|\cdot\|$ and inner product $\cdot^\top$ are assumed to represent the Euclidean norm and metric, respectively, unless explicitly specified otherwise. Non-Euclidean inner products are represented by the notation $\langle \cdot, \cdot \rangle_\gamma$, where γ represents the underlying metric, while norms are represented $\|\cdot\|_\gamma$. Throughout the paper, the Frobenius norm and inner product are used in conjunction with matrix algebras.

2.2 Path Assumptions

A path refers to a parametrized regular function $\phi : [0, 1] \rightarrow \mathbb{R}^n$, $\phi \in C^\infty$ with bounded curvature¹ that gives the target curve in Cartesian coordinates for any point along the normalized domain. Define $\phi_1(t)$ to be the first component of this vector, $\phi_2(t)$ to be the second component, and so forth. We write the unit tangent vector to ϕ at t as $T_\phi(t)$, and the unit normal vector as $N_\phi(t)$. Since $\mathcal{P} := \text{Im}(\phi)$ is a 1-dimensional submanifold of $\mathbb{R}^n$, the tangent space is well-defined and contains only one vector for any t . Let the smooth function $\theta : [0, 1] \rightarrow \text{SO}(n)$, $\theta \in C^\infty$ give the target heading orientations of the robot as it follows the path, where $\text{SO}(n)$ refers to the special orthogonal group of degree n . We refer to the robot’s target heading over the interval $[0, 1]$, as given by θ , as a heading interpolation. In particular, for every $t \in [0, 1]$, we assign this t a unique reference position along the curve given by $\phi(t)$ and target heading value $\theta(t)$.

We proceed under the assumption that the robot operates on the workspace $\mathbb{R}^n$, under which the inner product of any two vectors is assumed to be the standard dot product unless otherwise explicitly stated. We equip the workspace with a function $\psi : \mathbb{R}^n \rightarrow [0, 1]$ that partitions $\mathbb{R}^n$ into level sets, mapping a point $p \in \mathbb{R}^n$ to the parametric t -value of the closest point on ϕ to p . The robot’s configuration space will be the manifold $\text{SE}(n)$, defined as the principal $\text{SO}(n)$ bundle over $\mathbb{R}^n$, such that $\text{SE}(n) \simeq \mathbb{R}^n \rtimes \text{SO}(n)$. Note that for our GVF, we augment the robot’s pose with twist, forming a state space of $T\text{SE}(n)$. We will assume obstacles are negligible for simplicity.

2.3 Robot Assumptions

This algorithm targets holonomic robots, that may have non-uniform velocity constraints in each direction of motion. Let $v_1^{\max}, v_2^{\max}, \dots, v_n^{\max}$ be the robot’s maximum velocities in each of the basis directions $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ in the robot’s frame of reference. We represent this information as an ellipsoid of velocities, $\mathcal{E}_v$, defined as follows:

$$\mathcal{E}_v := \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{A} \mathbf{v} \leq 1\}, \mathbf{A} = \text{diag}\left(\frac{1}{(v_1^{\max})^2}, \frac{1}{1/(v_2^{\max})^2}, \dots, \frac{1}{(v_n^{\max})^2}\right)$$

In particular, we can conclude the existence of the following maximum velocity bound:

$$v_{\max} := \max_{1 \leq i \leq n} v_i^{\max} \geq \|\mathbf{v}\| \quad (1)$$

¹Note that the Pedro Pathing library can follow any paths or heading interpolations that are C^1 with bounded curvature, but we impose C^∞ smoothness to disregard exotic behavior for proofs in this paper.

where $\mathbf{v}$ represents the robot's velocity vector at any given time. Additionally, assume the robot converts the output of the GVF algorithm into a voltage vector $\mathbf{u}$ supplied to the actuators driving the motion. The actuators' control input is assumed to be uniquely determined by the velocity and acceleration, using the model $\mathbf{u} = \mathbf{u}(\mathbf{v}, \dot{\mathbf{v}})$, where u_i represents the control input supplied to actuator i , and therefore defining the field on second-order dynamics outputs a set of voltage inputs supplied to the actuators.

3 Guiding Vector Field

In this section we introduce some of the key geometric properties of our path-following algorithm. We assume the robot desires to follow a path $\Gamma \subset \text{SE}(n)$, a compact submanifold with boundary, parameterized with the position and rotation functions $\phi(t)$ and $\theta(t)$ on the interval $[0, 1]$. In particular we have the definition $\Gamma := \{(\phi(t), \theta(t)) \in \text{SE}(n) : t \in [0, 1]\}$. Our GVF will define second-order path-following dynamics, mapping a robot's pose and twist to a desired acceleration field. We refer to the vector field as $\xi : T\text{SE}(n) \rightarrow T(T\text{SE}(n))$, with a component-wise form $\xi(\mathbf{x}) = (\mathbf{v}, R\Omega, \xi_P(\mathbf{x}), \xi_R(\mathbf{x}))$, with $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega)$. This section will begin by defining some vector field preliminaries, and then future sections will tackle individual components of the vector field.

3.1 Closest Point Computation

A key component of this path-following algorithm is the ability to compute the closest point on the curve to the robot's current point. In this section, we make precise the algorithm to compute ψ for a given input. Let the robot's target curve be defined as ϕ , and the robot's current configuration $\mathbf{p}$. Then our goal is to minimize the squared distance function $D(t) = \frac{1}{2} \|\phi(t) - \mathbf{p}\|^2$. Note that this minimum occurs either at a critical point of D , or at the boundary points $t = 0$ and $t = 1$. Differentiating D , we find that $D'(t) = (\phi(t) - \mathbf{p}) \cdot \phi'(t)$. When the closest point is not uniquely defined, we choose the point that has the closest parametric t value to the robot's previous estimate for ψ . A root-finding algorithm can save time using the robot's previous known t -value as the initial-guess to begin the search process. For small sampling periods, the robot's previous t -value should be in close proximity to its current t -value.

3.2 Translational Correction

Translational correction attracts the robot towards the curve, allowing for the integral curves to approach and converge to the GVF. We choose some proportional and derivative gains $k_p^t, k_d^t > 0$ to control the magnitude of the vectors in the field, where the superscripts are used to avoid ambiguity with the other vector fields. The translational error can be defined by the distance from the robot to the curve ϕ . Using this controller to define the magnitude of our translational correction field, we assign to each $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in T\text{SE}(n)$, the following vector, for some small scalar $\epsilon_t > 0$:

$$\xi_{1T}(\mathbf{x}) = \left(k_p^t - k_d^t \frac{\mathbf{e}_{\mathbf{p}}^\top \mathbf{v}}{\|\mathbf{e}_{\mathbf{p}}\|^2 + \epsilon_t} \right) \mathbf{e}_{\mathbf{p}}, \quad \mathbf{e}_{\mathbf{p}} = \phi(\psi(\mathbf{p})) - \mathbf{p} \quad (2)$$

The field given by ξ_T operates in the direction of the shortest path from the robot to the path.

3.3 Heading Correction

Heading correction follows a similar principle to translational correction, allowing the robot to correct its orientation to match the target heading interpolation θ . This allows the robot to turn while driving along a path, while also holding and correcting its heading in case of interference from outside forces. Choose scalar proportional and derivative gains $k_p^r, k_d^r, k_{d2}^r > 0$ to minimize the norm of the robot's rotational error $r \in \mathfrak{so}(n)$, by driving the state R_1 to R_2 . To complete this definition, we equip the Lie algebra $\mathfrak{so}(n)$ with the Frobenius inner product, $\langle \cdot, \cdot \rangle = \text{tr}(\cdot^\top \cdot)$, and its corresponding norm. The rotational error can be computed by producing the relative rotation matrix from the robot's current orientation to the target orientation defined by θ , and then using the skew-projection operator to map the result to $\mathfrak{so}(n)$. We also augment this with a term that tracks the acceleration of the reference heading interpolation θ . In order to accomplish this, we define the function $\Omega_d(t) : [0, 1] \rightarrow \mathfrak{so}(n)$ that gives the necessary body angular velocity to track θ :

$$\Omega_d(t) = \theta(t)^\top \dot{\theta}(t), \quad \dot{\Omega}_d(t) = \dot{\theta}^\top \dot{\theta} + \theta^\top \ddot{\theta} \quad (3)$$

By assumption, $\dot{\Omega}_d(1) = 0$ in order to enforce that the robot does not expect the need for angular acceleration at steady state, when it has reached the end of the path. Using this definition, we define the rotational field ξ_R , for each $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$ as:

$$\xi_{1R}(\mathbf{x}) = \left(k_p^r - k_d^r \frac{\langle \mathbf{e}_R, \Omega \rangle_F}{\|\mathbf{e}_R\|_F^2 + \epsilon_r} \right) \mathbf{e}_R + k_{d2}^r \dot{\Omega}_d(\psi(\mathbf{p})), \quad \mathbf{e}_R = \mathbb{P}_a(\theta(\psi(\mathbf{p}))R^\top) \quad (4)$$

for some small scalar $\epsilon_r > 0$, where $\mathbb{P}_a$ denotes the standard skew-projection operator, $\mathbb{P}_a(X) = \frac{1}{2}(X - X^\top)$.

3.4 Centripetal Force

Another component of ξ is the centripetal force field. Generating centripetal force allows the robot to follow high-curvature curves more accurately, staying on the path for the duration of path-following. We can apply a simple feedforward loop on the robot's centripetal force, the corresponding force to this unwanted acceleration component, in order to generate enough centripetal force to follow ϕ . At a given parametric value $t \in [0, 1]$, the robot's required centripetal force can be described as follows, where m is the mass of the robot and v represents the robot's tangential speed:

$$\mathbf{F}_c = mv^2 \kappa N = mv^2 \frac{\|T'_\phi(t)\|}{\|\phi'(t)\|} \cdot N_\phi(t) = mv^2 N_\phi(t)$$

In practice, centripetal force generation can be implemented well using a scaling factor k_c , which absorbs parameters such as mass. For any $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$, we assign the centripetal force vector that corresponds to the closest point on the path to $\mathbf{p}$, thereby mapping all points in the same equivalence class under ψ to the same value in $\mathbb{R}^n$. Our final centripetal force field can therefore be defined as follows:

$$\xi_{1C}(\mathbf{x}) = k_c \left(\text{comp}_{T_\phi(\psi(\mathbf{p}))}(\mathbf{v}) \right)^2 \frac{T'_\phi(\psi(\mathbf{p}))}{\|\phi'(\psi(\mathbf{p}))\|} \quad (5)$$

From this point forward, we assume this vector has been tuned to provide exactly the correct amount of centripetal acceleration required for the robot to follow ϕ .

3.5 Tangential Drive

The tangential drive component is the main force that drives the robot along the path, and therefore has the highest magnitude of the four components. As expected, this vector acts in the direction of $T_\phi(t)$. We wish to integrate smooth deceleration into this vector, allowing the robot to brake cleanly at the end of the path. In order to accomplish this, we can use kinematic modeling to predict the ideal drive velocity given the distance remaining along the path. Choose a proportional gain $k_p^d > 0$ that supplies the necessary acceleration to drive the robot's current velocity to the target tangential velocity. For any $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$, we can compute the distance remaining along the path using the following equation:

$$d = \int_{\psi(\mathbf{p})}^1 \|\phi'(s)\| ds \quad (6)$$

Let a be the positive scalar representing the magnitude of the robot's available deceleration. Let $T = T_\phi(\psi(\mathbf{p}))$ be the current unit tangent vector to ϕ , $N = N_\phi(\psi(\mathbf{p}))$ be the current unit normal vector, $\mathbf{v}_\parallel$ be the robot's current tangential velocity obtained by projecting the robot's measured velocity onto T , and $v_\parallel = \|\mathbf{v}_\parallel\|$ be the robot's current tangential speed. We write the initial speed goal to stop deceleration as $v_{\text{desired}} = \sqrt{2ad}$ using basic kinematic manipulation². Finally, we define the drive velocity error to be $\mathbf{e}_D = |v_{\text{desired}} - v|$. Feeding the resulting error to our controller, we obtain a tangential drive vector of

$$\xi_{1D}(\mathbf{x}) = k_p^d \mathbf{e}_D T_\phi(\psi(\mathbf{p})), \quad \mathbf{e}_D = \left| \sqrt{2ad} - \text{comp}_{T_\phi(\psi(\mathbf{p}))}(\mathbf{v}) \right| \quad (7)$$

²External forces, such as friction, gravity, and drag, often alter the necessary initial velocity to produce a desired braking distance. The Pedro Pathing library fits the robot to a model $d(\mathbf{v}_\parallel)$, giving the robot's braking distance in terms of the initial velocity, and inverts this function to find the appropriate initial velocity. This also allows for introducing directional dependence in a .

3.6 Smooth Extension

The final GVF is therefore defined as follows, for $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$:

$$\xi_1(\mathbf{x}) = (\mathbf{v}, R\Omega, \xi_{1P}(\mathbf{x}), \xi_{1R}(\mathbf{x})) = (\mathbf{v}, R\Omega, \xi_T(\mathbf{x}) + \xi_C(\mathbf{x}) + \xi_D(\mathbf{x}), \xi_R(\mathbf{x})) \quad (8)$$

Note that ξ is not smooth on all of $TSE(n)$, since it makes use of the closest point function $\psi(x)$. In order to retain smoothness on all of $TSE(n)$, we modified ξ using a blend. The Tubular Neighborhood Theorem [Lee13] guarantees that ψ is smooth on any tubular neighborhood around ϕ . In this section, we extend the definition of ξ to be smooth on all of $TSE(n)$.

Let R_t be the minimum tubular radius of ϕ at all points³. Since ϕ has bounded curvature, this should be a finite value. Then any point $x \in \mathbb{R}^n$ within a Euclidean distance of R_t from ϕ will be in a tubular neighborhood of a point in ϕ . By the Tubular Neighborhood Theorem, the projection ψ will be well-defined and smooth for any point in this collection of points within a distance R_t of ϕ . Define a smooth cutoff $\eta : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ such that $\eta(r) = 1$ whenever $0 \leq r \leq 0.75 R_t$ and $\eta(r) = 0$ whenever $r \geq R_t$. We define the smooth bump $\chi : \mathbb{R}^n \rightarrow [0, 1]$, $x \mapsto \eta(\|x - \phi(\psi(x))\|)$. Let B be the smallest closed Euclidean ball containing ϕ , with center $c \in \mathbb{R}^n$ and radius R_B , and define a distance function $d(x) = \min_{y \in B} \|x - y\|$. We can represent d with the explicit formula $d(x) = \max\{0, \|x - c\| - R_B\}$, which shows that d is smooth everywhere except when $x \in \partial B$. Fix some small $\epsilon > 0$. Let $\tilde{\eta}$ be a scalar C^∞ mollifier satisfying the following for some constant C :

$$\tilde{\eta}(s) = \begin{cases} C \exp\left(-\frac{1}{1-(s/\epsilon)^2}\right) & |s| < \epsilon \\ 0 & |s| \geq \epsilon \end{cases}$$

$$\int_0^1 \tilde{\eta}(s) ds = 1$$

Now, we can define the smoothened function, using this mollifier and a modified version of our distance function:

$$q(x) = \int_0^1 \tilde{\eta}(s) \max\{0, \|x - c\| - R_B - s\} ds \quad (9)$$

Define the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as follows, for any $x \in \mathbb{R}^n$:

$$f(x) = \chi(x)\|x - \phi(\psi(x))\| + (1 - \chi(x))q(x) \quad (10)$$

It is trivial to verify that f is smooth, nonnegative, and has the zero level set ϕ . We define a new vector field using this definition, for when the robot's configuration $x \in SE(n)$ does not have a positional part that lies in a tubular neighborhood around ϕ . We will only use translational and tangential drive vector fields for this definition, since centripetal force proves to be less important at greater distances from the curve and rotational changes can be withheld until the robot returns to a tubular neighborhood of the curve. The new vector field for this case will be a standard GVF to follow the level set $f = 0$. Given $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$, we define the vector field as follows, for some small scalars $\epsilon_1, \epsilon_2, \epsilon_3 > 0$:

$$\xi_2(\mathbf{x}) = (\mathbf{v}, R\Omega, \xi_{2P}(\mathbf{x}), \xi_{2R}(\mathbf{x})) = \left(\mathbf{v}, R\Omega, k_p^d \tilde{\mathbf{e}}_{\mathbf{D}} w(\mathbf{p}) - k_T \frac{f(\mathbf{p}) \nabla f(\mathbf{p})}{\|\nabla f(\mathbf{p})\|^2 + \epsilon_1}, 0 \right), \quad (11)$$

$$w(\mathbf{p}) = \frac{\left(I - \frac{\nabla f(\mathbf{p}) \nabla f(\mathbf{p})^\top}{\|\nabla f(\mathbf{p})\|^2 + \epsilon_3} \right) T_\phi(0.5)}{\left\| \left(I - \frac{\nabla f(\mathbf{p}) \nabla f(\mathbf{p})^\top}{\|\nabla f(\mathbf{p})\|^2 + \epsilon_3} \right) T_\phi(0.5) \right\| + \epsilon_2}$$

where the definitions for k_p^d and $\tilde{\mathbf{e}}_{\mathbf{D}}$ are reused from (7), using the notation $\tilde{\mathbf{e}}_{\mathbf{D}}$ instead of $\mathbf{e}_{\mathbf{D}}$ to avoid ambiguity when combining the two vector fields, while the value $k_T > 0$ is a new proportional scalar gain. Note that in order to use the same definition of $\mathbf{e}_{\mathbf{D}}$, we need a way to quantify the remaining distance on the path. We use the following smooth approximation for the distance remaining on the curve, with respect to the robot's position $\mathbf{p}$:

$$S(\mathbf{p}) = \int_{\tilde{t}(\mathbf{p})}^1 \|\phi'(t)\| dt, \tilde{t}(\mathbf{p}) = \frac{\int_0^1 t e^{-\alpha \|\mathbf{p} - \phi(t)\|^2} dt}{\int_0^1 e^{-\alpha \|\mathbf{p} - \phi(t)\|^2} dt}$$

³In practice, we only need to choose R_t as the minimum of a neighborhood of points around the robot's predicted closest point on the path

This approximation approaches the actual path arc length as α approaches infinity, but this estimate does not need to be too precise since this calculation is used solely for deceleration. By the time the deceleration becomes heavily important, the robot should ideally already be within a tubular neighborhood of the path. With this definition, we can now define our final GVF for any $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n)$:

$$\boldsymbol{\xi}(\mathbf{x}) = (\mathbf{v}, R\Omega, \boldsymbol{\xi}_P(\mathbf{x}), \boldsymbol{\xi}_R(\mathbf{x})) = \chi(\mathbf{p})\boldsymbol{\xi}_1(\mathbf{x}) + (1 - \chi(\mathbf{p}))\boldsymbol{\xi}_2(\mathbf{x}) \quad (12)$$

where $\chi(\mathbf{p})$ is defined the same way as in the definition for $f(x)$. By our construction, this vector field is globally smooth on its domain in $TSE(n)$.

4 Analysis

We now study the properties of this GVF. We are particularly interested in the convergence of the robot's pose in $SE(n)$ to the reference path ϕ . We analyze the ordinary differential equation (ODE) system

$$\dot{\mathbf{x}} = \boldsymbol{\xi}(\mathbf{x}) \quad (13)$$

where $\mathbf{x} \in TSE(n)$.

Theorem 4.1. (*Path Invariance*) *For any $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in T\Gamma$, the condition $\dot{\Omega} = c\dot{\Omega}_d$ holds for some scalar $c > 0$, and $\dot{v}$ provides precisely enough normal acceleration to maintain path-following on ϕ .*

Proof. When $\mathbf{x} \in \Gamma$, we have that $\chi(\mathbf{p}) = 1$ and therefore $\boldsymbol{\xi}(\mathbf{x}) = \boldsymbol{\xi}_1(\mathbf{x})$. Note that the only position components of this vector field are $\boldsymbol{\xi}_{1C}$ and $\boldsymbol{\xi}_{1D}$, as the translational component vanishes as $\mathbf{e}_p \rightarrow 0$. With the centripetal force field providing precisely the required normal acceleration for the robot to follow ϕ , along with the drive vector field acting purely tangential to ϕ , their sum produces precisely the correct amount of centripetal acceleration to keep the robot on ϕ . Furthermore, as $\mathbf{e}_R$ vanishes, the angular acceleration becomes precisely $k_{d2}^r \dot{\Omega}_d$, as can be seen from (4). This allows the robot to track the reference heading interpolation θ as it drives along the path. These conditions together ensure that the robot's state remains in $T\Gamma$ post-convergence. $\square$

Theorem 4.2. (*Local Nondegeneracy*) *There exists a neighborhood $U \in TSE(n)$ around $T\Gamma$ such that $\mathbf{x}^* = (\phi(1), \theta(1), 0, 0)$ is precisely the only zero of the vector field $\boldsymbol{\xi}$.*

Proof. We begin by proving that $\mathbf{x}^*$ is a zero of $\boldsymbol{\xi}$. This guarantees that if the robot is at the target endpoint and has no velocity, the robot will also have no acceleration: an important property that allows the robot to remain at the endpoint indefinitely. Note that $\boldsymbol{\xi}(\mathbf{x}^*) = \boldsymbol{\xi}_1(\mathbf{x}^*)$ since $\chi(\mathbf{x}^*) = 1$. The translational component vanishes at Γ , while the heading error $\mathbf{e}_R \rightarrow 0$ and the rotational component converges to a constant times $\dot{\Omega}_d(1)$, which we know to be 0 by assumption. Additionally, as the remaining distance along the path approaches 0, the desired tangential speed approaches 0. With $v_{\text{desired}} = v = 0$ at the path's endpoint, we obtain $\mathbf{e}_D = 0$, and thus the tangential drive component vanishes. Finally, the centripetal force component scales off the robot's velocity, and thus also vanishes. Therefore, with only zero components at $\mathbf{x}^*$, the vector field $\boldsymbol{\xi}$ vanishes there. Choose the neighborhood $U := \{\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n) : \mathbf{p} \in A^\circ\}$, where $A = \chi^{-1}(1)$. We now proceed to show that for any $\mathbf{x} \in U, \mathbf{x} \neq \mathbf{x}^*$, the vector field $\boldsymbol{\xi}$ does not vanish: $\boldsymbol{\xi}(\mathbf{x}) \neq \mathbf{0}$. Assume for the sake of contradiction that $\boldsymbol{\xi}(\mathbf{x}) = \mathbf{0}$ for some $\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega)$ satisfying the conditions above. Since $\chi(\mathbf{p}) = 1$, we again have that $\boldsymbol{\xi}(\mathbf{x}) = \boldsymbol{\xi}_1(\mathbf{x})$. We restrict our attention to the position component $\boldsymbol{\xi}_{1P}$ of this vector field. In order for $\boldsymbol{\xi}(\mathbf{x})$ to be 0, the robot's velocity must also be the zero vector, which implies that the drive component cannot vanish as $\mathbf{v}_{\text{desired}} \neq 0$ and therefore $\mathbf{e}_D \neq 0$. By the Tubular Neighborhood Theorem, we know that $\mathbf{e}_p \cdot T_\phi(\psi(\mathbf{p})) = 0$, which implies that the translational vector field acts purely in the orthogonal direction, a property that also applies to the centripetal vector field. Therefore, $\boldsymbol{\xi}$ will always have a nonzero component of acceleration in the tangential direction, forcing a contradiction. Therefore, $\mathbf{x}^*$ is the only zero of $\boldsymbol{\xi}$ in the neighborhood⁴ U . $\square$

For the following proofs, we treat $\mathbf{x} = \mathbf{x}(t) = (\mathbf{p}(t), R(t), \mathbf{v}(t), \Omega(t))$ as a trajectory and analyze the convergence of this trajectory. First, define the distance function $s(t) := f(\mathbf{p}(t))$. Computing the first and second-order derivatives of s yields the following:

$$\dot{s}(t) = \nabla f(\mathbf{p}(t))^\top \mathbf{v}(t), \quad \ddot{s}(t) = \mathbf{v}(t)^\top \nabla^2 f(\mathbf{p}(t)) \mathbf{v}(t) + \nabla f(\mathbf{p}(t))^\top \dot{\mathbf{v}}(t) \quad (14)$$

This function serves as a distance function from the path ϕ . We begin by proving a simple result that validates the translational and heading controllers $\boldsymbol{\xi}_{1T}$ and $\boldsymbol{\xi}_{1R}$.

⁴A very similar argument would work to show that $\boldsymbol{\xi}$ does not vanish on the level set $\chi(\mathbf{p}) = 0$, but showing this property in the general case for all of $TSE(n)$ proves far more difficult.

Lemma 4.3. *The scalar gains k_p^t and k_p^r can be chosen to ensure the vector fields ξ_{1T} and ξ_{1R} act positively in the direction of $\mathbf{e}_p$ and $\mathbf{e}_R$, ignoring the contribution from the term proportional to $\dot{\Omega}_d$ for the heading field. The magnitude of this vector can be made as large as needed.*

Proof. A similar proof works for either vector field. We prove this for the translational vector field, and the rotational case follows from the analogous proof for the heading vector field, where each dot product is replaced with the Frobenius inner product. We wish to show that

$$k_p^t \geq k_d^t \frac{\mathbf{e}_p^\top \mathbf{v}}{\|\mathbf{e}_p\|^2 + \epsilon_t}$$

is a reasonable constraint, regardless of $\mathbf{p}$ and $\mathbf{v}$. Applying Cauchy-Schwarz and (1), we get the following bound:

$$k_d^t \frac{\mathbf{e}_p^\top \mathbf{v}}{\|\mathbf{e}_p\|^2 + \epsilon_t} \leq k_d^t \frac{\|\mathbf{e}_p\| v_{\max}}{\|\mathbf{e}_p\|^2 + \epsilon_t}$$

The right-hand-side of this equation is maximized at $\|\mathbf{e}_p\| = \sqrt{\epsilon_t}$, which can be verified by computing the critical points of $\|\mathbf{e}_p\|$. The result is the following loose⁵ inequality:

$$k_d^t \frac{\mathbf{e}_p^\top \mathbf{v}}{\|\mathbf{e}_p\|^2 + \epsilon_t} \leq k_d^t \frac{2}{\sqrt{\epsilon_t}}$$

Choosing k_p^t to be greater than the right-hand side of the equation allows the controller to maintain a positive contribution in the direction of $\mathbf{e}_p$, and k_p^t can be chosen to make this contribution as large as needed. $\square$

Lemma 4.4. *(Finite-Time Convergence to Inner Tube) Define the set $A := \{\mathbf{x} = (\mathbf{p}, R, \mathbf{v}, \Omega) \in TSE(n) : \chi(\mathbf{p}) < 1\}$. Furthermore, assume that $\|\nabla f\| \geq g_0$ for some scalar $g_0 > 0$ whenever $\mathbf{x} \in A$. Then, for sufficiently well-chosen scalar gains, any trajectories $\mathbf{x}(t)$ satisfying $\mathbf{x}(0) \in A$ leave A in finite time.*

Proof. Let $f_0 > 0$ be such that $f(\mathbf{p}) \geq f_0$ on A . This is justifiable since $\mathbf{p}$ is far from the path, and therefore f is relatively large on A . Furthermore, since f is smooth, let $h > 0$ be such that $\|\nabla^2 f\| \leq h$. Since the position dynamics are governed by ξ_{2P} on A , we can write the following expression,

$$\dot{\mathbf{v}} = \left(\left(k_p^t - k_d^t \frac{\mathbf{e}_p^\top \mathbf{v}}{\|\mathbf{e}_p\|^2 + \epsilon_t} \right) \mathbf{e}_p - k_p^d \mathbf{e}_D T_\phi(\psi(\mathbf{p})) \right) \chi(\mathbf{p}) + \left(k_p^d \tilde{\mathbf{e}}_D w(\mathbf{p}) - k_T \frac{f(\mathbf{p}) \nabla f(\mathbf{p})}{\|f(\mathbf{p})\|^2 + \epsilon_1} \right) (1 - \chi(\mathbf{p})),$$

with all expressions aligning with the definition in (11). We substitute our expression for $\dot{\mathbf{v}}$ into the expansion of $\ddot{s}$ from (14), omitting the tangential term since $\nabla f(\mathbf{p})^\top w(\mathbf{p}) = 0$, given that $w(p)$ was defined as the normalized projection of $T_\phi(0.5)$ onto the orthogonal complement of $\nabla f(\mathbf{p})$. Furthermore, working over tubular neighborhoods allows us to conclude that $\mathbf{e}_p$ acts in the same direction as $\nabla f(\mathbf{p})$, and in particular $\mathbf{e}_p = f(\mathbf{p}) \nabla f(\mathbf{p})$. Additionally, $\nabla f(\mathbf{p})^\top T_\phi(\mathbf{p}) = 0$, so the tangential drive contribution from ξ_1 also vanishes. The new expression for $\ddot{s}$ becomes

$$\ddot{s}(t) = \mathbf{v}^\top \nabla^2 f(\mathbf{p}) \mathbf{v} - k_T (1 - \chi(\mathbf{p})) f(\mathbf{p}) \frac{\|\nabla f(\mathbf{p})\|^2}{\|\nabla f(\mathbf{p})\|^2 + \epsilon_1} - \chi(\mathbf{p}) f(\mathbf{p}) \beta, \quad \beta := \left(k_p^t - k_d^t \frac{\mathbf{e}_p^\top \mathbf{v}}{\|\mathbf{e}_p\|^2 + \epsilon_t} \right)$$

If we substitute in our bounds on f and its derivatives, along with the velocity bound from (1), we obtain the following inequality:

$$\ddot{s}(t) \leq h v_{\max}^2 - ((1 - \chi(\mathbf{p})) k_T + \chi(\mathbf{p}) \beta) f_0 \frac{g_0^2}{g_0^2 + \epsilon_1}$$

Choose k_T and k_p^t large enough to satisfy the following condition, for an arbitrary $\epsilon_0 > 0$:

$$k_T \geq \gamma, \quad k_p^t \geq \gamma + k_d^t \frac{4R_t}{9R_t^2 + 16\epsilon}, \quad \gamma := \frac{(h v_{\max}^2 + \epsilon_0)(g_0^2 + \epsilon_1)}{f_0 g_0^2}$$

⁵In practice, this bound is much higher than necessary and k_p^t can therefore be chosen much less strictly than the restrictions imposed in this paper.

The second inequality guarantees that $\beta \geq \gamma$. In particular, substituting this condition into the inequality for $\ddot{s}$ gives the following set of inequalities:

$$\begin{aligned}\ddot{s}(t) &\leq hv_{\max}^2 - ((1 - \chi(\mathbf{p}))k_T + \chi(\mathbf{p})\beta)f_0 \frac{g_0^2}{g_0^2 + \epsilon_1} \leq hv_{\max}^2 - ((1 - \chi(\mathbf{p}))\gamma + \chi(\mathbf{p})\gamma)f_0 \frac{g_0^2}{g_0^2 + \epsilon_1} \\ &= hv_{\max}^2 - \gamma f_0 \frac{g_0^2}{g_0^2 + \epsilon_1} = -\epsilon_0\end{aligned}$$

Integrating each side of the expression $\ddot{s}(t) \leq -\epsilon_0$ twice yields the following inequality:

$$s(t) \leq s(0) + \dot{s}(0)t - \frac{1}{2}\epsilon_0 t^2$$

Here, if $\mathbf{x}(t)$ stays in A infinitely, this forms a contradiction because the right-hand-side of the inequality becomes negative for large t , whereas $s(t) := f(\mathbf{p}(t)) \geq 0$. Therefore, $\mathbf{x}(t)$ must leave A before the right-hand-side becomes negative, which occurs in finite time. $\square$

Lemma 4.5. (*Asymptotic Convergence Inside Inner Tube*) Assume the robot enters the level set $\chi^{-1}(1)$ at time T_1 . For sufficiently well-chosen scalar gains, the robot asymptotically converges to $T\Gamma$.

Proof. With $\chi(\mathbf{p}) = 1$, the robot's dynamics are governed by $\dot{\mathbf{x}} = \boldsymbol{\xi}_1(\mathbf{x})$. In particular, $\dot{\mathbf{v}} = \boldsymbol{\xi}_{1T} + \boldsymbol{\xi}_{1C} + \boldsymbol{\xi}_{1D}$. We decompose the robot's velocity in terms of the orthogonal and tangential components to the path, defining $\mathbf{v}_{\parallel} := \text{proj}_{T_{\phi}(\psi(\mathbf{p}))}(\mathbf{v})$ and $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$, where $\mathbf{v}_{\perp}$ can also be defined as $\text{proj}_{\mathbf{e}_{\mathbf{p}}}(\mathbf{v})$ when $\mathbf{e}_{\mathbf{p}} \neq \mathbf{0}$. Note that expanding $\dot{\mathbf{e}}_{\mathbf{p}}$ gives the following expression:

$$\dot{\mathbf{e}}_{\mathbf{p}} = \phi'(\psi(\mathbf{p}))\psi' - \mathbf{v} \quad (15)$$

This tells us that the normal component of $\dot{\mathbf{e}}_{\mathbf{p}}$ is essentially given by $\mathbf{v}_{\perp}$, a fact that will become helpful later.

If the time T_1 , when the robot enters the level set $\chi^{-1}(1)$, satisfies the condition $T_1 > 0$, the normal velocity $\mathbf{v}_{\perp}$ satisfies $\mathbf{e}_{\mathbf{p}}^{\top} \mathbf{v}_{\perp} > 0$ since the robot must have a velocity towards the path to enter this level set. On the other hand, if $T_1 = 0$ and the robot starts in the level set $\chi = 1$, there must be a time T_2 where the robot's position lies in the level set $\chi = 1$ and the velocity satisfies this condition. The proof of this statement is as follows: as long as $\mathbf{e}_{\mathbf{p}} > 0$, the robot must have an acceleration in the direction towards the path. This implies that the robot cannot retain the condition $\mathbf{v}_{\perp} = 0$ over a continuous period of time. If we assume that the robot begins with a velocity away from the path, and assume for the sake of contradiction that the robot maintains this velocity away from the path, then there must be some timestamp T_3 where the robot leaves the level set $\chi = 1$. By Lemma (4.4), the robot must return to the level set $\chi = 1$ and therefore hold a velocity toward the path, forming a contradiction. We now consider the system's evolution following the timestamp T_2 , where the robot's position is in the level set $\chi = 1$ and $\mathbf{e}_{\mathbf{p}}^{\top} \mathbf{v}_{\perp} > 0$.

We now consider the direction of the robot's normal velocity vector $\mathbf{v}_{\perp}$. Because $\boldsymbol{\xi}_{1D}$ has no contribution to $\mathbf{v}_{\perp}$, and $\boldsymbol{\xi}_{1C}$ was explicitly defined only to cancel curvature terms accumulating in $\dot{\mathbf{v}}$, only the translational terms are critical to analyze the change in the contribution of $\mathbf{v}$ in the direction orthogonal to ϕ . From Lemma (4.3), we know that scalar gains for $\boldsymbol{\xi}_{1T}$ can be chosen to ensure that $\mathbf{e}_{\mathbf{p}}^{\top} = c\boldsymbol{\xi}_{1T}(\mathbf{x})$ for a scalar $c > 0$, and therefore $\mathbf{e}_{\mathbf{p}}^{\top} \boldsymbol{\xi}_1(\mathbf{x}) = \mathbf{e}_{\mathbf{p}}^{\top} \dot{\mathbf{v}} > 0$ under the assumption that $\mathbf{e}_{\mathbf{p}} > 0$. Since $\boldsymbol{\xi}_{1T}$ acts in the direction of $\mathbf{e}_{\mathbf{p}}$, the normal velocity must also continue to act in the direction of $\mathbf{e}_{\mathbf{p}}$. Similarly, $\boldsymbol{\xi}_{1C}$ was designed to cancel out the tangential contribution to $\mathbf{e}_{\mathbf{p}}$, allowing us to consider only the normal component of $\dot{\mathbf{e}}_{\mathbf{p}}$. Since $\mathbf{v}_{\perp}$ acts towards ϕ , we can conclude that

$$\frac{d}{dt} \|\mathbf{e}_{\mathbf{p}}\| < 0 \quad (16)$$

We now define the following Lyapunov candidate function:

$$V = \frac{1}{2} \|\mathbf{e}_{\mathbf{p}}\|^2$$

Clearly $V \geq 0$, with equality holding precisely when $\mathbf{p}(t) \in \mathcal{P}$. Note that the time-derivative of V is negative, using (16). Furthermore, $V \rightarrow \infty$ as $\|\psi(\mathbf{p}(t)) - \mathbf{p}(t)\| \rightarrow \infty$, and $R(t)$ is bounded by the compactness of $\text{SO}(n)$, so trajectories must stay bounded. Importantly, with $\dot{V} < 0$, the robot does not get further from the path and therefore leave the set $\chi = 1$. Applying Lyapunov's Direct Method [Kha02], we conclude that $\mathbf{e}_{\mathbf{p}} \rightarrow \mathbf{0}$ as $t \rightarrow \infty$.

Furthermore, since $\dot{\mathbf{v}}$ depends directly on $\boldsymbol{\xi}_1$, the robot's velocity is bounded, the path ϕ is smooth, and the robot's position is confined to the compact level set $\chi = 1$, we know that $\ddot{\mathbf{e}}_{\mathbf{p}}$ is strictly bounded. This implies that

$\dot{\mathbf{e}}_{\mathbf{p}}$ is uniformly continuous, and, by a corollary of Barbalat's Lemma [Kha02], we can conclude that $\dot{\mathbf{e}}_{\mathbf{p}} \rightarrow \mathbf{0}$. Since (15) tells us that $\mathbf{v}_{\perp}$ is essentially the normal component of $\dot{\mathbf{e}}_{\mathbf{p}}$, the convergence $\dot{\mathbf{e}}_{\mathbf{p}} \rightarrow \mathbf{0}$ implies that $\mathbf{v}_{\perp} \rightarrow \mathbf{0}$. This means that $\mathbf{v}$ converges to the tangent bundle $T\mathcal{P}$. We now prove that the rotational error $\mathbf{e}_{\mathbf{R}} \rightarrow \mathbf{0}$. In order to continue working on flat Euclidean space, we map our rotational tangent vectors from the Lie algebra $\mathfrak{so}(n)$ to $\mathbb{R}^{n(n-1)/2}$ using the vee operator. Observe that we can rewrite the rotational dynamics of ξ_1 as follows:

$$\begin{aligned}\dot{\mathbf{e}}_{\mathbf{R}}^{\vee} &= (\Omega_d - \Omega)^{\vee} \\ \ddot{\mathbf{e}}_{\mathbf{R}}^{\vee} &= (\dot{\Omega}_d - \dot{\Omega})^{\vee} = -\left(k_p^r - k_d^r \frac{\langle \mathbf{e}_{\mathbf{R}}, \Omega \rangle_F}{\|\mathbf{e}_{\mathbf{R}}\|^2 + \epsilon_R}\right) \mathbf{e}_{\mathbf{R}}^{\vee} - k_{d2}^r \dot{\Omega}_d^{\vee} + \dot{\Omega}_d^{\vee}\end{aligned}$$

Cleaning up the second expression yields the following ordinary differential equation:

$$\ddot{\mathbf{e}}_{\mathbf{R}}^{\vee} = -c(t) \mathbf{e}_{\mathbf{R}}^{\vee} + (1 - k_{d2}^r) \dot{\Omega}_d^{\vee}, c(t) := \left(k_p^r - k_d^r \frac{\langle \mathbf{e}_{\mathbf{R}}, \Omega \rangle_F}{\|\mathbf{e}_{\mathbf{R}}\|^2 + \epsilon_R}\right) \quad (17)$$

Note that since θ is smooth, it must have bounded first and second derivatives. Expanding $\dot{\Omega}_d^{\vee}$ yields the following expression:

$$\dot{\Omega}_d = \ddot{\theta} \left(\frac{d\psi}{dt}\right)^2 + \dot{\theta} \frac{d^2\psi}{dt^2}$$

This implies that $\|\dot{\Omega}_d^{\vee}\|$ must be bounded from above by some constant A . The last equation defines an autonomous input-to-state stable dissipative system. Furthermore, from (4.3), we can force a lower bound $0 < c_{\min} < c(t)$. The asymptotic behavior of the second-order differential equation satisfies the following bound:

$$\lim_{t \rightarrow \infty} \|\mathbf{e}_{\mathbf{R}}^{\vee}\| \leq \frac{|1 - k_{d2}^r|A}{c_{\min}}$$

and therefore choosing scalar gains to force the lower bound $c_{\min}$ larger or driving $|1 - k_{d2}^r|$ smaller forces $\mathbf{e}_{\mathbf{R}}^{\vee} \rightarrow \mathbf{0}$ to any required precision. This implies that the robot's heading R practically converges to θ . Taking the norm of both sides of (17) and applying the triangle inequality yields:

$$\|\ddot{\mathbf{e}}_{\mathbf{R}}\| \leq c(t) \|\mathbf{e}_{\mathbf{R}}^{\vee}\| + |1 - k_{d2}^r|A$$

Analyzing the convergence behavior as $t \rightarrow \infty$, we can substitute our bounds on the limit of $\|\mathbf{e}_{\mathbf{R}}^{\vee}\|$:

$$\lim_{t \rightarrow \infty} \|\ddot{\mathbf{e}}_{\mathbf{R}}\| \leq |1 - k_{d2}^r|A \left(\frac{c_{\max}}{c_{\min}} + 1\right)$$

Applying Landau's inequality allows us to bound the asymptotic limit of $\dot{\mathbf{e}}_{\mathbf{R}}$ based on our bounds for the limits of $\mathbf{e}_{\mathbf{R}}$ and $\ddot{\mathbf{e}}_{\mathbf{R}}$:

$$\lim_{t \rightarrow \infty} \|\dot{\mathbf{e}}_{\mathbf{R}}\| \leq 2|1 - k_{d2}^r|A \sqrt{\frac{1}{c_{\min}} \left(\frac{c_{\max}}{c_{\min}} + 1\right)}$$

For simplicity, define the vectors $\boldsymbol{\omega}_d = \Omega_d^{\vee}$ and $\boldsymbol{\omega} = \Omega^{\vee}$. Furthermore, we define $\hat{\boldsymbol{\omega}}_d$ as the normalized form of $\boldsymbol{\omega}_d$. We define the orthogonal projection matrix $P_{\hat{\boldsymbol{\omega}}_d}^{\perp} = I - \hat{\boldsymbol{\omega}}_d (\hat{\boldsymbol{\omega}}_d)^{\top}$. The length of the component of $\boldsymbol{\omega}_{\perp}$ orthogonal to the reference angular velocity $\hat{\boldsymbol{\omega}}_d$ can be written as follows, using the fact that $P_{\hat{\boldsymbol{\omega}}_d}^{\perp} \hat{\boldsymbol{\omega}}_d = \mathbf{0}$:

$$\|P_{\hat{\boldsymbol{\omega}}_d}^{\perp} \boldsymbol{\omega}\| = \|P_{\hat{\boldsymbol{\omega}}_d}^{\perp} (\hat{\boldsymbol{\omega}}_d - \boldsymbol{\omega})\| = \|P_{\hat{\boldsymbol{\omega}}_d}^{\perp} \dot{\mathbf{e}}_{\mathbf{R}}\|$$

Since orthogonal projection is a non-expansive operation, the following result is a direct consequence of the previous equality:

$$\|P_{\hat{\boldsymbol{\omega}}_d}^{\perp} \boldsymbol{\omega}\| \leq \|\dot{\mathbf{e}}_{\mathbf{R}}\|$$

Analyzing the convergence of either side of the limit, we can finally bound the robot's deviation from the reference angular velocity:

$$\lim_{t \rightarrow \infty} \|P_{\hat{\boldsymbol{\omega}}_d}^{\perp} \boldsymbol{\omega}\| \leq \lim_{t \rightarrow \infty} \|\dot{\mathbf{e}}_{\mathbf{R}}\| \leq 2|1 - k_{d2}^r|A \sqrt{\frac{1}{c_{\min}} \left(\frac{c_{\max}}{c_{\min}} + 1\right)} \quad (18)$$

Similarly to the practical convergence of the robot's heading to θ , the convergence of the robot's angular velocity can be forced to converge to the tangent bundle $T(\text{Im}(\theta))$ to any degree of precision either by minimizing $|1 - k_{d2}^r|$ or by increasing $c_{\min}$. This concludes our proof that all four components of the robot's state can eventually be made to converge to their reference targets, and thus $\mathbf{x}(t)$ converges to $T\Gamma$ asymptotically. $\square$

We now summarize these results with the theorem presented below:

Theorem 4.6. (*Global Asymptotic Convergence*) For any initial conditions $\mathbf{x}(0) \in TSE(n)$, with sufficiently well-chosen scalar gains for ξ , the robot asymptotically converges to the tangent bundle TT as $t \rightarrow \infty$.

Proof. This follows immediately from Lemma (4.4) and Lemma (4.5). $\square$

5 Conclusion

This paper presented a novel dimension-agnostic guiding vector field-based path-following controller. Building the controller from four essential components, the paper showed how the vector field’s smoothness could be extended to the entire ambient space, despite the difficulties posed by following parametric representations of curves. Defining the state along the tangent bundle of the configuration manifold, the paper also extended classical GVF techniques from primarily first-order systems to second-order control, allowing for the robustness of geometric path-following to be used for robots relying on acceleration-level commands. Through rigorous mathematical justification, the paper also showed that the tangent space of the reference path serves as an invariant set of the GVF system, and that, regardless of initial conditions, the system finds asymptotically stable convergence to the reference path to any desired degree of precision.

The structural stability and geometric guarantees of the framework allow the algorithm to serve as a robust and lightweight controller for practical use. The framework serves as the basis of the Pedro Pathing algorithm, which successfully utilizes this algorithm in the space $SE(2)$ to power thousands of robots across the world. Future work will explore the integration of obstacle avoidance methodologies along with extensions for compatibility with underactuated systems to handle non-holonomic constraints.

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